

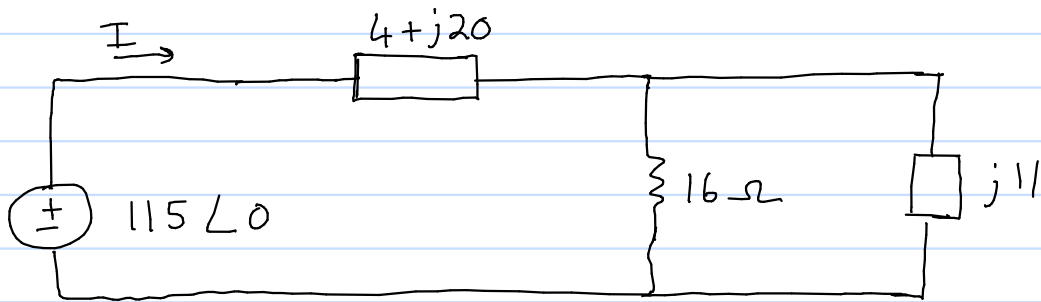
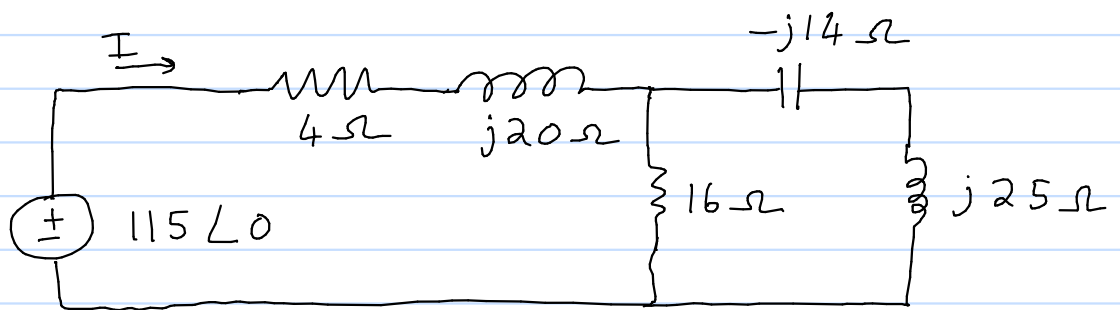
9.28) Determine the current that flows through an $20\text{-}\Omega$ resistor connected to a voltage source $V_s = 120\cos(377t + 37^\circ)\text{ V}$

$$i(t) = \frac{V(t)}{R}$$

$$= \frac{120\cos(377t + 37^\circ)}{20}$$

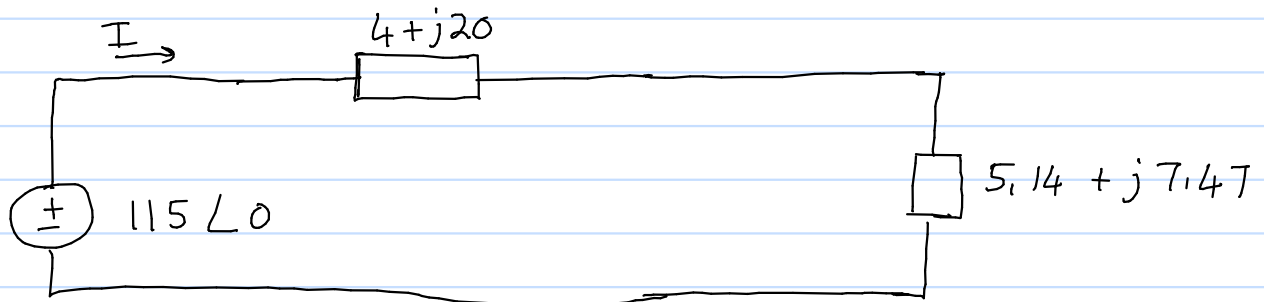
$$= 6\cos(377t + 37^\circ)\text{ A}$$

9.39) Find Z_{eq} and I in the circuit:



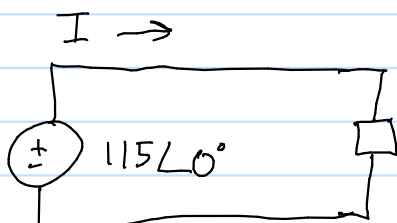
Series combinations:

$$\begin{aligned} 4 + j20 & \quad (1) \\ -j14 + j25 & \end{aligned}$$



parallel combination

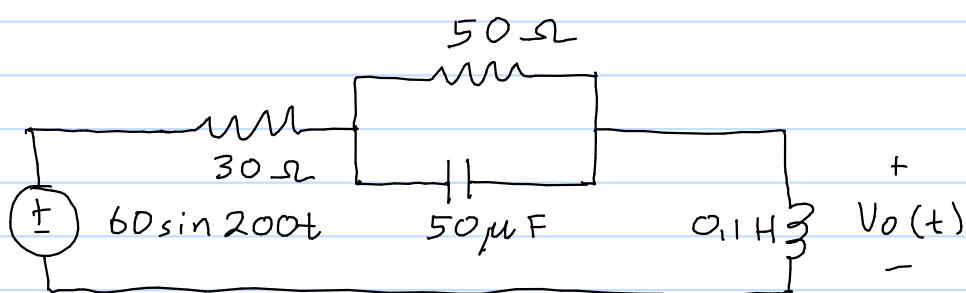
$$\begin{aligned} 16 \parallel (j11) &= \frac{16 \times j11}{16 + j11} \\ &= \frac{j176}{16 + j11} \cdot \frac{(16 - j11)}{(16 - j11)} \\ &= \frac{1936 + j2816}{377} \end{aligned}$$



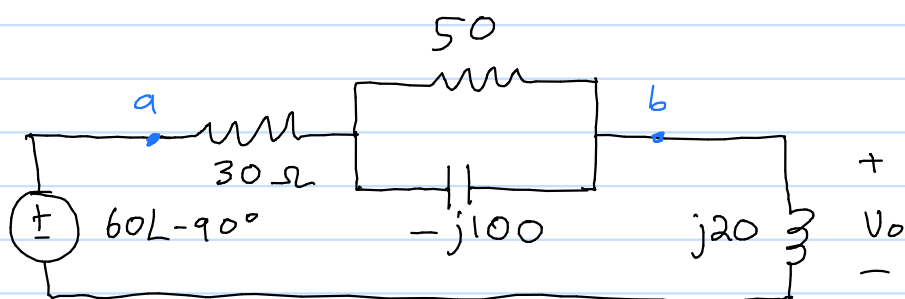
$$\begin{aligned} Z_{eq} &= 5.14 + j7.47 + 4 + j20 \\ &= 9.14 + j27.47 \end{aligned}$$

$$\begin{aligned} I &= \frac{115 \angle 0^\circ}{9.14 + j27.47} \\ &= \frac{115 \angle 0^\circ}{28.95 \angle 71.60^\circ} \\ &= 3.97 \angle -71.60^\circ \end{aligned}$$

9.42) Calculate $V_o(t)$ in the circuit



Convert everything to phasors. $\omega = 200 \text{ rad/s}$

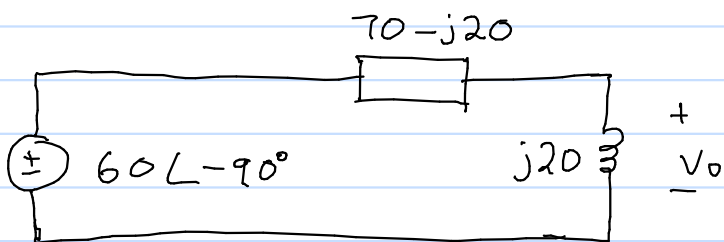


$$50 \mu\text{F} \Rightarrow \frac{1}{j200 \times 50 \times 10^{-6}} = -j100$$

$$0.1 \text{ H} \Rightarrow j200 \times 0.1 = j20$$

$$60 \sin 200t \Rightarrow 60 \angle (0 - 90) = 60 \angle -90^\circ$$

Find the equivalent resistance between a-b



$$Z_{eq} = 50 \parallel (-j100) + 30$$

$$= \frac{50 \times (-j100)}{50 - j100} \times \frac{(50 + j100)}{(50 + j100)} + 30$$

$$= \frac{500000 - j250000}{12500} + 30$$

$$= 40 - j20 + 30$$

$$= 70 - j20$$

Using voltage division:

$$V_o = \frac{60 \angle -90 \times j20}{j20 + 70 - j20}$$

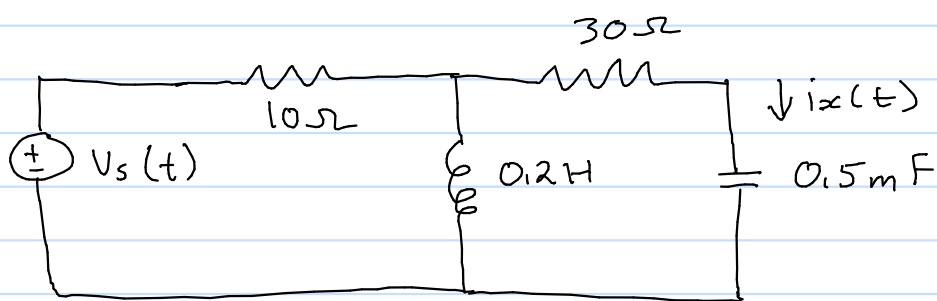
$$= \frac{60 \angle -90 \times 20 \angle 90}{70}$$

$$= \frac{120 \angle 0^\circ}{7} = 17.14$$

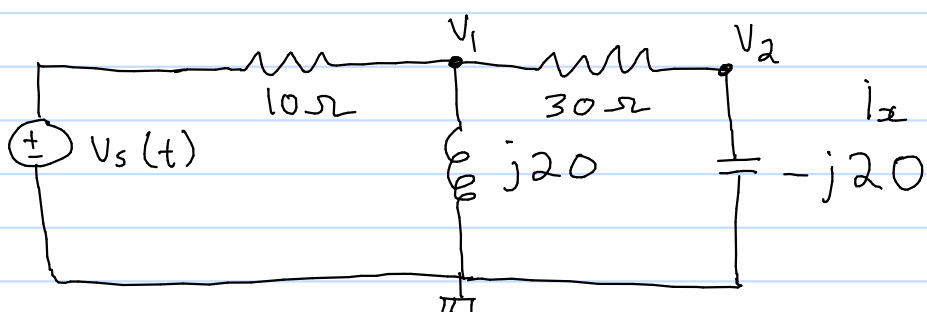
Convert back to time domain:

$$V_o(t) = 17.14 \cos(200t)$$

9.48) Given $V_s(t) = 200 \sin(100t - 40^\circ)$
determine $i_x(t)$.



Convert to phasors.



$$200 \sin(100t - 40^\circ) \Rightarrow 200 \angle (-40^\circ - 90^\circ) = 200 \angle -130^\circ$$

$$0.2 \text{ H} \Rightarrow j100 \times 0.2 = j20 \text{ ohms}$$

$$0.5 \text{ mF} \Rightarrow \frac{1}{j100 \times 0.0005} = -j20$$

KCL at V_1 :

$$\frac{200 \angle -130^\circ - V_1}{10} = \frac{V_1}{j20} + \frac{V_1}{30 - j20}$$

$$20 \angle -130^\circ - \frac{V_1}{10} = \frac{V_1}{j20} + \frac{V_1}{30 - j20}$$

$$V_1 \left(\frac{1}{j20} + \frac{1}{30 - j20} + \frac{1}{10} \right) = 20 \angle -130^\circ$$

$$V_1 = 20 \angle -130^\circ \left(\frac{1}{j20} + \frac{1}{30 - j20} + \frac{1}{10} \right)^{-1}$$

$$= 20 \angle -130^\circ \left(\frac{-j}{20} + \frac{30 + j20}{1300} + \frac{1}{10} \right)^{-1}$$

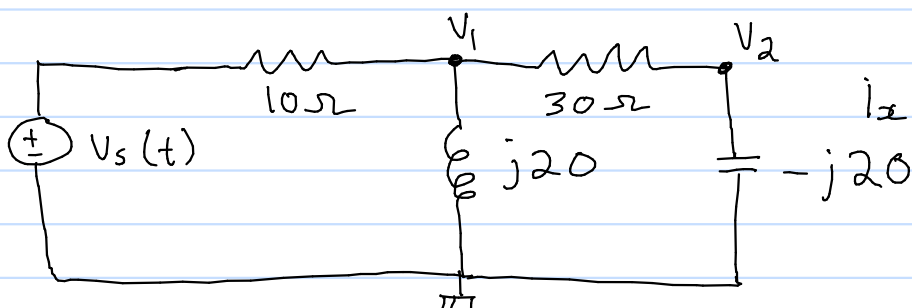
$$= 20 \angle -130^\circ \left(\frac{-j65 + 30 + j20 + 130}{1300} \right)^{-1}$$

$$= 20 \angle -130^\circ \left(\frac{160 - j45}{1300} \right)^{-1}$$

$$= 20 \angle -130^\circ (0.128 \angle -15.709^\circ)^{-1}$$

$$= \frac{20 \angle -130^\circ}{0.128 \angle -15.709^\circ}$$

$$V_1 = 156.25 \angle -114.291^\circ$$



Ohm's law:

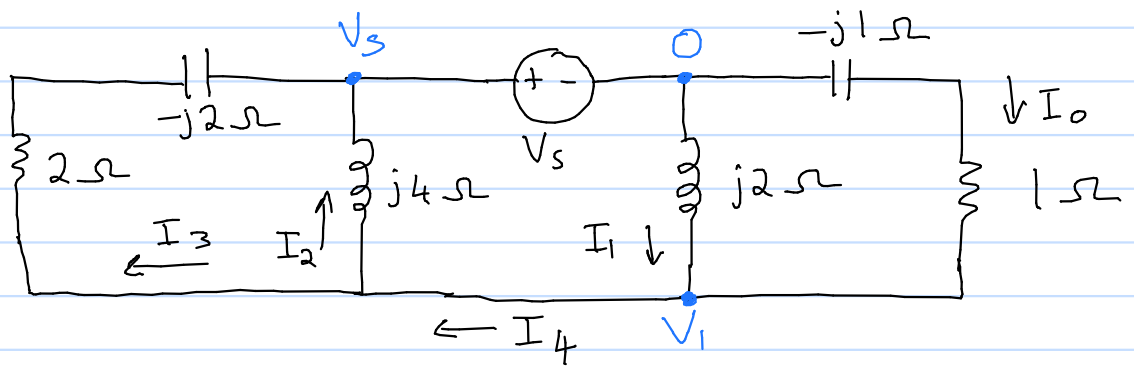
$$i_x = \frac{V_1}{30 - j20}$$

$$= \frac{156.25 \angle -114.291^\circ}{36.056 \angle -33.69^\circ}$$

$$= 4.333 \angle -80.601^\circ$$

$$i_x(t) = 4.333 \cos(100t - 80.601^\circ)$$

9.54) find V_s if $I_o = 30 \angle 0^\circ \text{ A}$



$$V_1 = -I_o(1 - j1) \quad \text{Ohm's law}$$

$$V_1 = -30 + j30$$

$$I_1 = -\frac{V_1}{j2} \quad \text{Ohm's law}$$

$$= \frac{+30}{j2} - \frac{j30}{j2}$$

$$= -15 - j15$$

$$I_4 = I_o + I_1 \quad \text{KCL}$$

$$= 30 - 15 - j15$$

$$= 15 - j15$$

$$I_2 = \frac{I_4 \times (2 - j2)}{2 - j2 + j4} \quad \text{Current division}$$

$$= \frac{(15 - j15)(2 - j2)}{2 + j2}$$

$$= \frac{(15 - j15)(2 - j2)^2}{(2 + j2)(2 - j2)}$$

$$= \frac{(15 - j15)(-j8)}{8}$$

$$= -15 - j15$$

$$I_2 = \frac{V_1 - V_s}{j4} \quad \text{Ohm's law}$$

$$(-15 - j15)j4 = V_1 - V_s$$

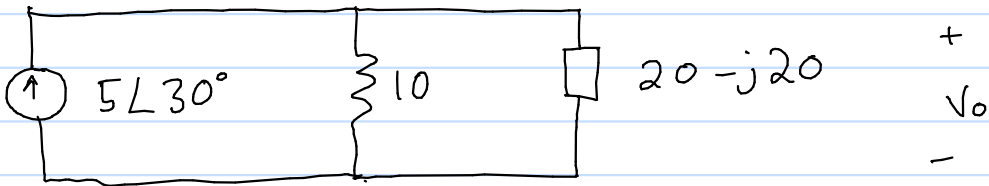
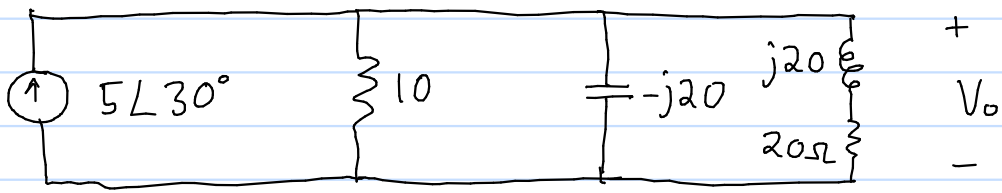
$$V_s = V_1 - (60 - j60)$$

$$V_s = -30 + j30 - 60 + j60$$

$$V_s = -90 + j90$$

$$V_s = 127.279 \angle 135^\circ$$

9.64) Find Z_T and V_o in the circuit



$$Z = (-j20) \parallel (j20 + 20)$$

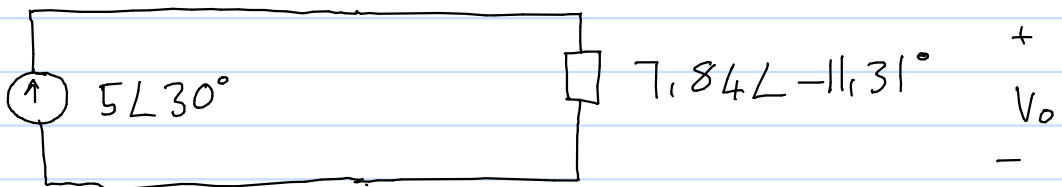
$$= \left(\frac{1}{-j20} + \frac{1}{20 + j20} \right)^{-1}$$

$$= \left(\frac{j}{20} + \frac{20 - j20}{800} \right)^{-1}$$

$$= \left(\frac{j40 + 20 - j20}{800} \right)^{-1}$$

$$= \frac{800}{20 + j20} = \frac{800(20 - j20)}{800}$$

$$= 20 - j20$$



$$Z_T = 10 \parallel (20 - j20)$$

$$= \left(\frac{1}{10} + \frac{1}{20 - j20} \right)^{-1}$$

$$= \left(\frac{2 - j2 + 1}{20 - j20} \right)^{-1}$$

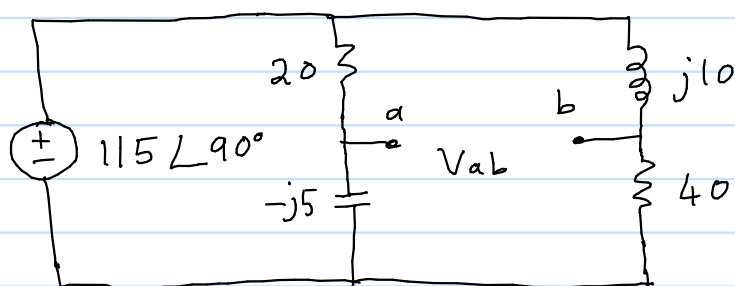
$$= \frac{20 - j20}{3 - j2} \times \frac{(3 + j2)}{(3 + j2)}$$

$$Z_T = \frac{100 - j20}{13} = 7.84 \angle -11.31^\circ$$

$$V_o = 5 \angle 30^\circ \times 7.84 \angle -11.31^\circ \quad \text{Ohm's law}$$

$$= 39.2 \angle 18.69^\circ$$

9.66) calculate Z_t and V_{ab}



$$V_a = \frac{115 \angle 90^\circ \times (-j5)}{20 - j5} \times \frac{(20 + j5)}{20 + j5} \quad \text{Voltage division}$$

$$= \frac{115 \angle 90^\circ \times (25 - 100j)}{425}$$

$$= \frac{115 \angle 90^\circ \times 103.08 \angle -75.96^\circ}{425}$$

$$= 27.89 \angle 14.04^\circ$$

$$V_b = \frac{115 \angle 90^\circ \times 40}{40 + j10} \times \frac{(40 - j10)}{40 - j10} \quad \text{Voltage division}$$

$$= \frac{115 \angle 90^\circ \times (1600 - j400)}{1700}$$

$$= \frac{115 \angle 90^\circ \times 1649.24 \angle -14.04^\circ}{1700}$$

$$= 111.57 \angle 75.96^\circ$$

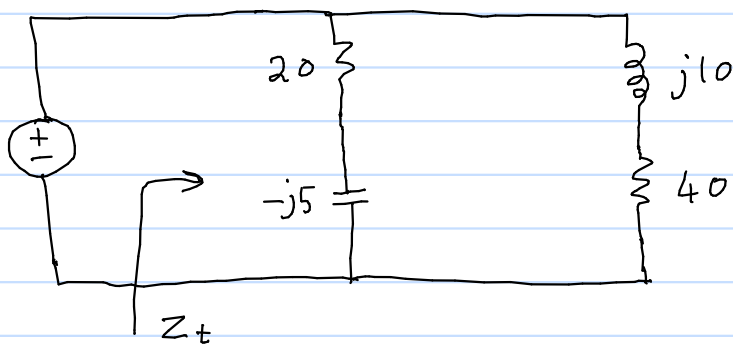
$$V_{ab} = V_a - V_b$$

$$= 27.89 \angle 14.04^\circ - 111.57 \angle 75.96^\circ$$

$$= 27.04 + j6.77 - 27.07 - j108.24$$

$$= -0.03 - j101.47$$

$$= 101.47 \angle -90.02^\circ$$



$$Z_t = (20 - j5) \parallel (40 + j10)$$

$$= \frac{(20 - j5)(40 + j10)}{20 + 40 - j5 + j10}$$

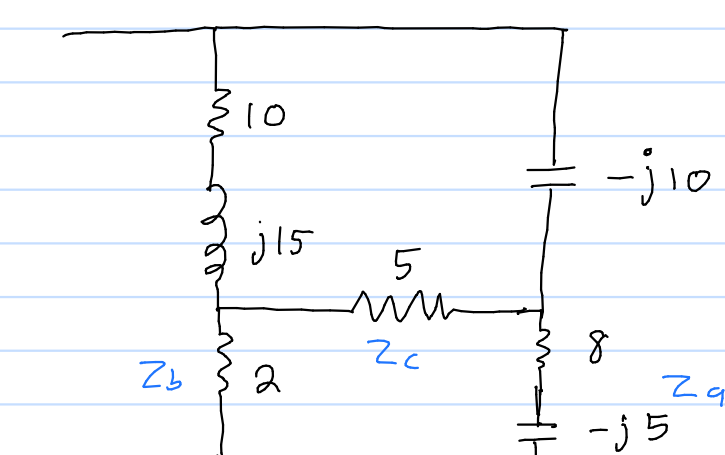
$$= \frac{850}{60 + j5}$$

$$= \frac{850}{60.208 \angle 4.764^\circ}$$

$$= 14.118 \angle -4.764^\circ$$

$$= 14.069 - j1.173$$

9.70) find the equivalent impedance



Δ -Y conversion

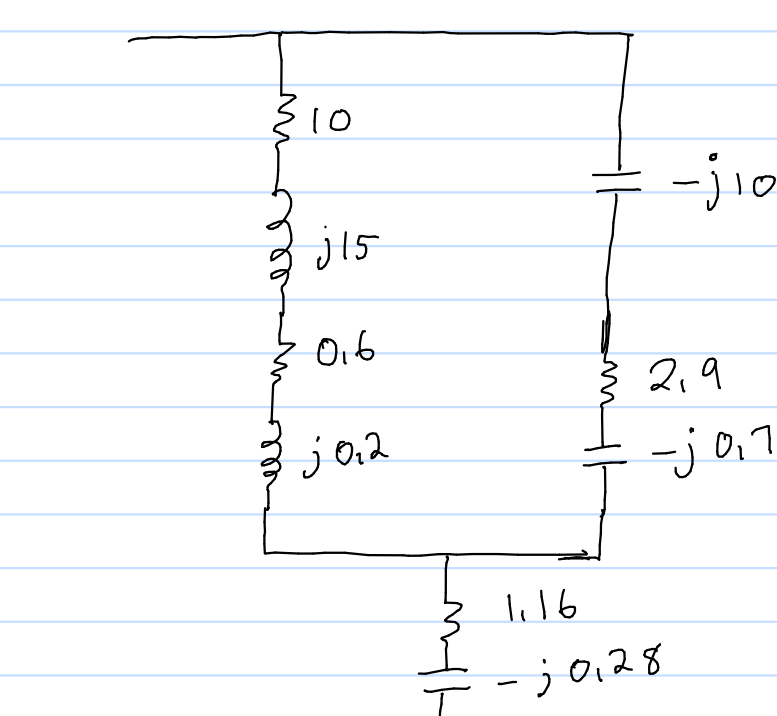
$$\begin{aligned} Z_t &= Z_a + Z_b + Z_c \\ &= 2 + 5 + 8 - j5 \\ &= 15 - j5 \end{aligned}$$

$$\begin{aligned} \frac{1}{Z_t} &= \frac{1}{15 - j5} \times \frac{(15 + j5)}{(15 + j5)} \\ &= \frac{15 + j5}{250} \end{aligned}$$

$$\begin{aligned} Z_1 &= \frac{Z_b Z_c}{Z_t} \\ &= \frac{2 \times 5 \times (15 + j5)}{250} \\ &= 0.6 + j0.2 \end{aligned}$$

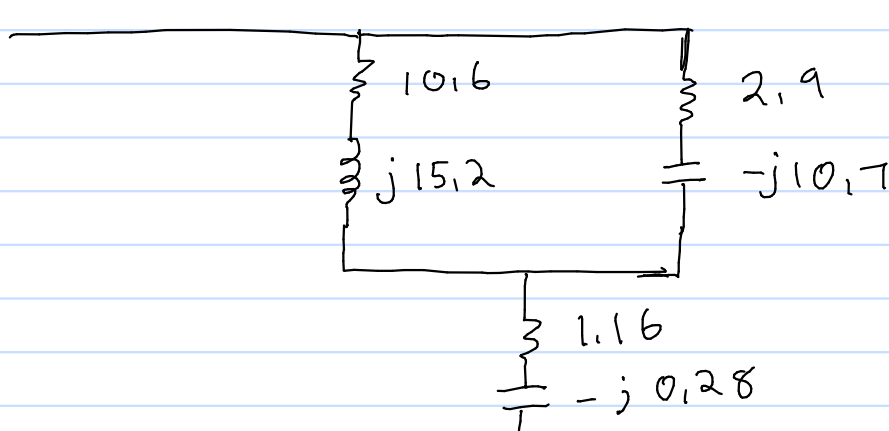
$$\begin{aligned} Z_2 &= \frac{Z_c Z_a}{Z_t} \\ &= \frac{5 \times (8 - j5)(15 + j5)}{250} \\ &= \frac{725 - j175}{250} \\ &= 2.9 - j0.7 \end{aligned}$$

$$\begin{aligned} Z_3 &= \frac{Z_b Z_a}{Z_t} \\ &= \frac{2 \times (8 - j5)(15 + j5)}{250} \\ &= \frac{290 - j70}{250} \\ &= 1.16 - j0.28 \end{aligned}$$



$$Z_1 = 10 + j15 + 0.6 + j0.2 = 10.6 + j15.2$$

$$Z_2 = 2.9 - j0.7 - j10 = 2.9 - j10.7$$

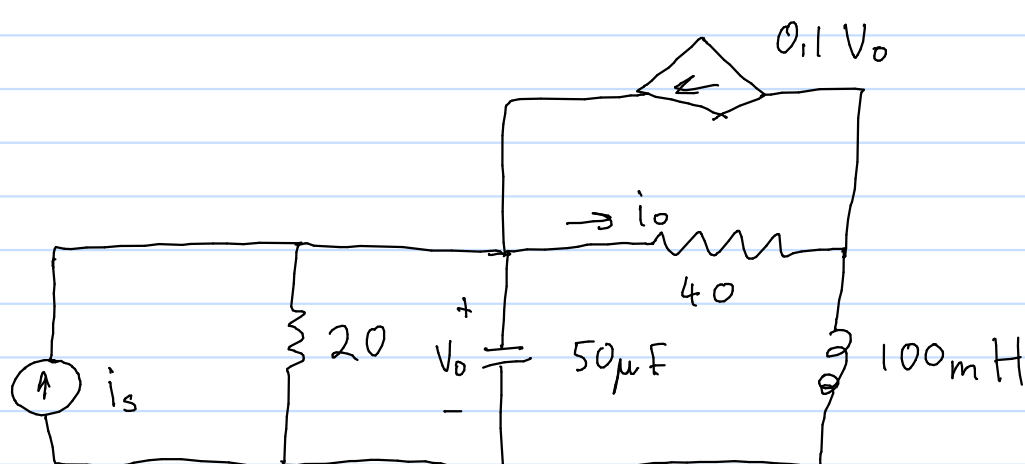


$$\begin{aligned} Z_1 || Z_2 &= \frac{(10.6 + j15.2)(2.9 - j10.7)}{10.6 + j15.2 + 2.9 - j10.7} \\ &= \frac{193.38 - j69.34}{13.5 + j4.5} \\ &= \frac{205.436 \angle -19.726^\circ}{14.23 \angle 18.435^\circ} \\ &= 14.437 \angle -38.161^\circ \\ &= 11.351 - j8.920 \end{aligned}$$

$$\begin{aligned} Z_{eq} &= 11.351 - j8.920 + 1.16 - j0.28 \\ &= 12.511 - j9.2 \end{aligned}$$

10.8) Use nodal analysis to find i_o if

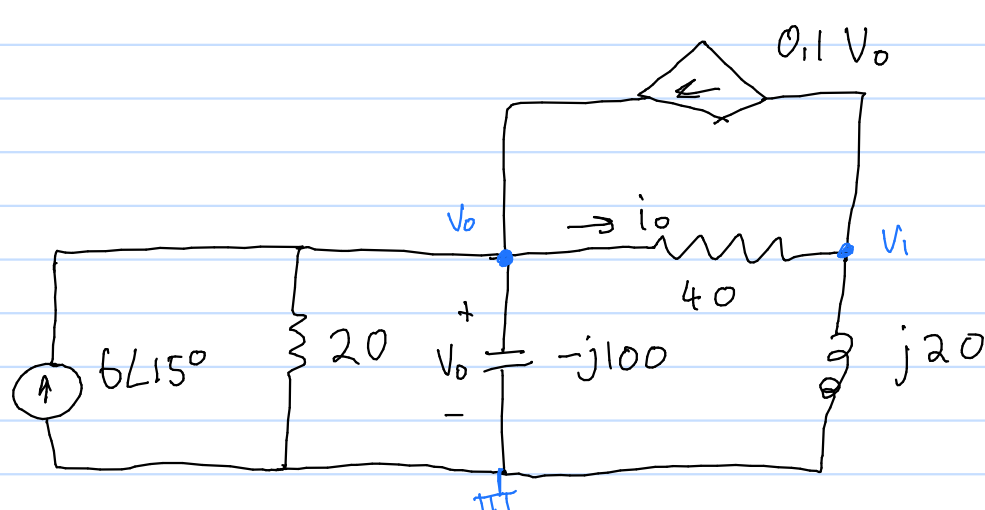
$$i_s = 6 \cos(200t + 15^\circ) \text{ A}$$



$$6 \cos(200t + 15^\circ) \Rightarrow 6 \angle 15^\circ$$

$$50 \mu\text{F} \Rightarrow \frac{1}{j200 \times 50 \times 10^{-6}} = -j100$$

$$100 \text{ mH} \Rightarrow j200 \times 100 \times 10^{-3} = j20$$



KCL at node V_o :

$$-6 \angle 15^\circ + \frac{V_o}{20} + \frac{V_o}{-j100} + \frac{V_o - V_1}{40} - 0.1 V_o = 0$$

$$V_o \left(\frac{1}{20} + \frac{j}{100} + \frac{1}{40} - \frac{1}{10} \right) - V_1 \left(\frac{1}{40} \right) = 6 \angle 15^\circ$$

$$V_o \left(\frac{-1}{40} + \frac{j}{100} \right) - V_1 \left(\frac{1}{40} \right) = 6 \angle 15^\circ$$

KCL at node V_1 :

$$\frac{V_1}{j20} + \frac{V_1 - V_o}{40} + 0.1 V_o = 0$$

$$V_o \left(\frac{-1}{40} + \frac{1}{10} \right) + V_1 \left(\frac{1}{j20} + \frac{1}{40} \right) = 0$$

$$V_o \left(\frac{3}{40} \right) + V_1 \left(\frac{1}{40} - \frac{j}{20} \right) = 0$$

Solve complex matrix equation:

$$\begin{bmatrix} -0.025 + j0.01 & -0.025 \\ 0.075 & 0.025 - j0.05 \end{bmatrix} \begin{bmatrix} V_o \\ V_1 \end{bmatrix} = \begin{bmatrix} 6 \angle 15^\circ \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2.5 + j & -2.5 \\ 7.5 & 2.5 - j5 \end{bmatrix} \begin{bmatrix} V_o \\ V_1 \end{bmatrix} = \begin{bmatrix} 600 \angle 15^\circ \\ 0 \end{bmatrix}$$

A

$$|A| = (-2.5 + j)(2.5 - j5) - (-2.5)(7.5) = 17.5 + j15$$

$$|A_o| = \begin{vmatrix} 600 \angle 15^\circ & -2.5 \\ 0 & 2.5 - j5 \end{vmatrix} = 600 \angle 15^\circ \times (2.5 - j5) = 2225.35 - j2509.55$$

$$|A_1| = \begin{vmatrix} -2.5 + j & 600 \angle 15^\circ \\ 7.5 & 0 \end{vmatrix} = -600 \angle 15^\circ \times 7.5 = -4346.67 - j1164.69$$

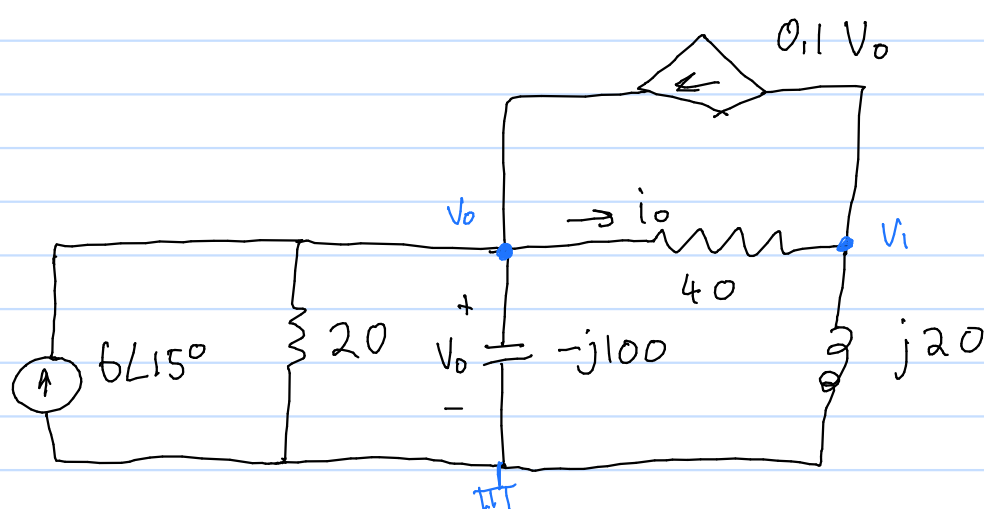
Using Cramer's rule:

$$V_o = \frac{|A_o|}{|A|} = \frac{2225.35 - j2509.55}{17.5 + j15}$$

$$= 2.448 - j145.5$$

$$V_1 = \frac{|A_1|}{|A|} = \frac{-4346.67 - j1164.69}{17.5 + j15}$$

$$= -176.07 + j84.36$$



$$i_o = \frac{V_o - V_1}{40}$$

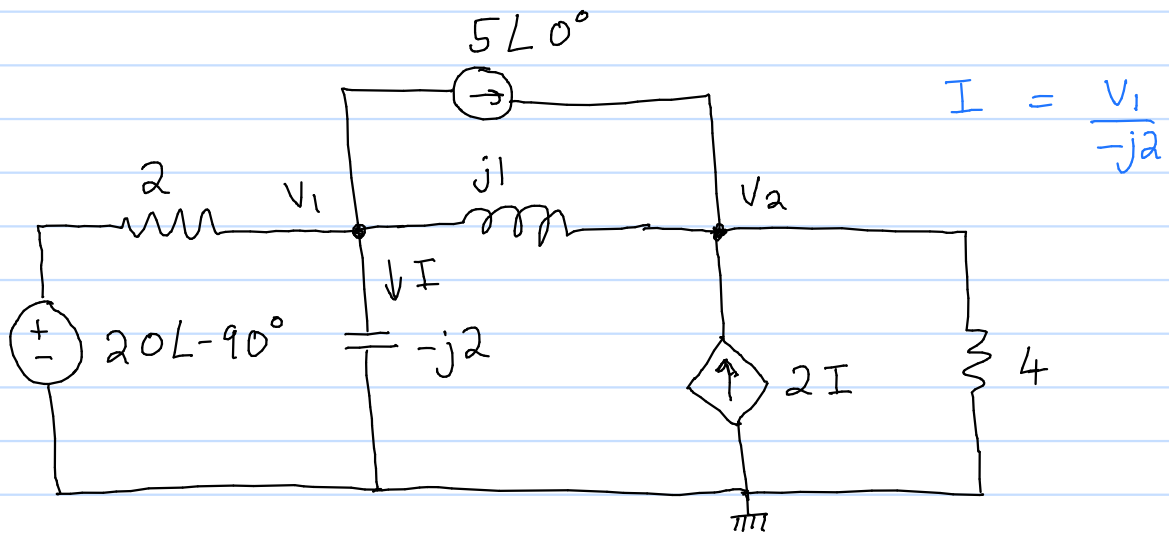
$$= \frac{2.448 - j145.5 + 176.07 - j84.36}{40}$$

$$= 4.463 - j5.747$$

$$= 7.276 \angle -52.166^\circ$$

$$i_o(t) = 7.276 \cos(200t - 52.166^\circ) \text{ A}$$

10.15) Solve for the current I using nodal analysis.



KCL at node 1:

$$\frac{V_1 - 20\angle -90^\circ}{2} + \frac{V_1}{-j2} + \frac{V_1 - V_2}{j} + 5 = 0$$

$$\frac{V_1}{2} + \frac{j20}{2} + \frac{V_1 j}{2} - V_1 j + V_2 j + 5 = 0$$

$$V_1 \left(\frac{1}{2} + \frac{j}{2} - j \right) + V_2 (j) = -5 - \frac{j20}{2}$$

$$V_1 (0.5 - j0.5) + V_2 j = -5 - j10$$

KCL at node 2:

$$\frac{V_2 - V_1}{j} - 5 + \frac{V_2}{4} - 2 \left(\frac{V_1}{-j2} \right) = 0$$

$$-V_2 j + V_1 j - 5 + \frac{V_2}{4} - V_1 j = 0$$

$$V_1 (j - j) + V_2 \left(\frac{1}{4} - j \right) = 5$$

$$V_2 (0.25 - j) = 5$$

$$V_2 = \frac{5}{0.25 - j}$$

$$V_2 = \frac{5(0.25 + j)}{1.0625}$$

$$V_2 = \frac{20 + j80}{17}$$

Substitute V_2 into e.g. 1:

$$V_1 (0.5 - j0.5) + V_2 j = -5 - j10$$

$$V_1 (0.5 - j0.5) + \frac{-80 + j20}{17} = -5 - j10$$

$$V_1 (0.5 - j0.5) = -5 - j10 + \frac{80 - j20}{17}$$

$$V_1 (0.5 - j0.5) = \frac{-5 - j190}{17}$$

$$V_1 = \frac{-5 - j190}{8.5 - j8.5}$$

$$V_1 = 10.882 - j11.471$$

$$I_1 = \frac{V_1}{-j2}$$

$$= \frac{10.882 - j11.471}{-j2}$$

$$= 5.735 + j5.441$$